

TE 454 - Digital Signal Processing

124 Multiple-Choice Questions with Answers

TE 454 Digital Signal Processing

1. In the context of this course, the phrase "digital signal processing" is best understood as:

- A. The branch of electronics devoted to designing analog amplifiers and filters
- B. The study of how physical sensors convert measured quantities into voltages
- C. The theories, methods and algorithms for processing signals in digital form ✓**
- D. The fabrication of microprocessor chips from raw semiconductor wafer material

Answer: C. The theories, methods and algorithms for processing signals in digital form

2. The term "digital signal processor" (as opposed to "digital signal processing") refers to:

- A. A kind of microprocessor used to implement digital signal processing algorithms ✓**
- B. The mathematical algorithm through which an input signal is passed for its result
- C. The analog front-end that band-limits a signal before it reaches the converter
- D. The high-level software language in which the processing routines are written

Answer: A. A kind of microprocessor used to implement digital signal processing algorithms

3. An analog signal is best characterised as one that is:

- A. Continuous in both time and magnitude ✓**
- B. Continuous in time but discrete in magnitude
- C. Discrete in time but continuous in magnitude
- D. Discrete in both time and magnitude

Answer: A. Continuous in both time and magnitude

4. A digital signal is best characterised as one that is:

- A. Discrete in both time and magnitude ✓**
- B. Continuous in both time and magnitude
- C. Continuous in time but discrete in magnitude
- D. Discrete in time but continuous in magnitude

Answer: A. Discrete in both time and magnitude

5. Before a real-world analog signal can be processed digitally, two transformations must be applied in order. They are:

- A. Quantize the signal in amplitude first, then sample it at discrete time points
- B. Sample the signal at discrete time points, then quantize the sample values ✓**
- C. Amplify the signal to full scale, then shift its frequency into a lower band
- D. Filter the signal to remove noise, then store it directly in continuous form

Answer: B. Sample the signal at discrete time points, then quantize the sample values

6. Quantization of a sampled signal is normally carried out by:

- A. Delaying each sample value by one sampling interval before it is stored
- B. Averaging each sample value together with its two nearest neighbours
- C. Rounding or truncating each sample value to an available level ✓**
- D. Multiplying each sample value by a fixed scaling constant before storage

Answer: C. Rounding or truncating each sample value to an available level

7. Why must most real-world signals be transformed before a DSP system can operate on them?

- A. Most real-world signals are already digital and merely need to be amplified
- B. Most real-world signals are analog, while DSP works on signals in digital form ✓**
- C. DSP systems can handle periodic signals but reject aperiodic ones outright

D. DSP systems require a signal to be expressed in the frequency domain first

Answer: B. Most real-world signals are analog, while DSP works on signals in digital form

8. Which statement best captures the relationship between digitization and DSP that the course uses to motivate the subject?

A. Digitization and DSP are competing techniques for handling the same signals

B. DSP is a narrow niche that has little bearing on the modern information technology field

C. Digitization underpins information technology, and its kernel is digital signal processing ✓

D. Digitization replaced DSP once integrated circuits became widely available

Answer: C. Digitization underpins information technology, and its kernel is digital signal processing

9. A key reason a digital system offers greater programmability than an analog system is that:

A. A digital system needs no software at all once its hardware has been fixed

B. A digital system rewires its own internal circuits as its requirements change

C. An analog system is reconfigured purely by editing its onboard firmware code

D. A digital system can be changed by modifying software rather than hardware ✓

Answer: D. A digital system can be changed by modifying software rather than hardware

10. An engineer swaps a standard filter for an adaptive filter simply by loading new code, leaving the hardware untouched. This illustrates which advantage of digital systems?

A. Programmability ✓

B. Anti-noise performance

C. Large-scale integration

D. Real-time response

Answer: A. Programmability

11. In a digital system, the precision of the processing is chiefly governed by:

A. The tolerance ratings printed on the resistors and the capacitors used

B. The number of ADC bits, the processor word-length and the algorithm used ✓

C. The ambient temperature and the humidity of the operating environment

D. The physical length of the wiring that connects the components together

Answer: B. The number of ADC bits, the processor word-length and the algorithm used

12. Why is an analog system's precision considered harder to guarantee than a digital system's?

A. Analog precision depends on component tolerances, such as resistors within 5% or capacitors within 20% ✓

B. Analog precision is fixed by the bit count of its analog-to-digital converter alone

C. Analog components retain their exact printed rated values throughout their whole service life

D. Analog systems round each sample coarsely, discarding the finer detail of the signal

Answer: A. Analog precision depends on component tolerances, such as resistors within 5% or capacitors within 20%

13. With respect to stability, how does a digital system typically compare with an analog one as temperature and humidity vary?

A. The digital system drifts more because its logic gates are temperature-sensitive

B. The digital system shows essentially no variation across its rated operating range ✓

C. Both systems drift by roughly the same amount for a given change in temperature

D. The digital system grows unstable while the analog system holds its values steady

Answer: B. The digital system shows essentially no variation across its rated operating range

14. The characteristics of analog components such as resistors, capacitors and op-amps are a concern for stability because they:

A. Change with temperature, humidity and similar environmental conditions ✓

B. Grow hard to measure accurately once they are soldered onto a circuit board

C. Consume far more electrical power than the equivalent digital logic gates do

D. Can be readjusted afterwards by rewriting the controlling software routines

Answer: A. Change with temperature, humidity and similar environmental conditions

15. The repeatability advantage of digital systems refers to the property that:

- A. A single digital unit can process many different inputs at the same time
- B. A digital signal can be repeated indefinitely without needing any storage
- C. Digital units detect and correct a wrong result soon after producing it

D. Identical digital units process the same input into the same output ✓

Answer: D. Identical digital units process the same input into the same output

16. Error-correcting codes are cited as an advantage of digital representation because they:

- A. Compress the data so that fewer bits need to be sent over the channel
- B. Scramble the data so that unauthorised receivers are unable to read it
- C. Convert the data back into an analog form that is suited to transmission

D. Add redundant bits so that introduced errors can be detected and corrected ✓

Answer: D. Add redundant bits so that introduced errors can be detected and corrected

17. Regarding fidelity in storage and transmission, the slides argue that:

- A. The fidelity of the analog medium exceeds that of the digital medium
- B. Digital and analog media deliver essentially identical fidelity in practice
- C. The fidelity of the digital medium is greater than that of the analog medium ✓
- D. Fidelity is governed mainly by the length of the transmission cable used

Answer: C. The fidelity of the digital medium is greater than that of the analog medium

18. How does the course contrast analog and digital data compression?

- A. Analog compression is the genuinely lossless kind, whereas digital compression loses information
- B. Neither analog nor digital compression is able to preserve the full information
- C. Analog compression loses some information, whereas digital technology makes lossless compression possible ✓
- D. Digital compression suits images well but is not applicable to audio signals

Answer: C. Analog compression loses some information, whereas digital technology makes lossless compression possible

19. The 3 kHz bandwidth limit applied to analog telephone lines is given as an example of:

- A. A lossless digital compression scheme applied to the line
- B. An error-correcting code operating on the telephone line
- C. Analog compression in which some information is lost ✓
- D. A sampling operation carried out before the quantization step

Answer: C. Analog compression in which some information is lost

20. Which is listed as a genuine limitation of digital signal processing rather than an advantage?

- A. Digital systems drift heavily whenever the operating temperature changes
- B. A/D and D/A conversion rates are not available for some applications, such as beyond GHz ✓
- C. Digital systems resist any reprogramming once they have been deployed out in the field
- D. Digital systems are unable to store their data for later retrieval and use

Answer: B. A/D and D/A conversion rates are not available for some applications, such as beyond GHz

21. One reason analog processing is still needed is real-time performance, because in an analog system:

- A. The processing time is set by the processor speed and the chosen algorithm
- B. Aside from circuit delay, the processing happens in real time ✓
- C. The signal must first be sampled before any output can begin to appear
- D. Each result must wait for a redundant error-correction pass to finish first

Answer: B. Aside from circuit delay, the processing happens in real time

22. Why can analog systems still be preferred for very high-frequency signals such as microwave or light-wave signals?

- A. Analog systems can store such high-frequency signals far more compactly than digital ones
- B. Such signals are limited by sample-and-hold, ADC and processor speeds, and fast ADCs cost a lot ✓**
- C. Very high-frequency signals carry no useful information that is worth the effort of digitizing
- D. Digital systems are designed to reject any signal lying above the standard audio band by rule

Answer: B. Such signals are limited by sample-and-hold, ADC and processor speeds, and fast ADCs cost a lot

23. The remark that "most signals in the real world are analog" is used in the course to argue that:

- A. Analog signals are best kept in analog form rather than converted into digital form
- B. DSP systems have by now made analog processing very nearly obsolete in practice
- C. Real-world signals arrive already sampled by nature and so need no conversion step
- D. Mixed-signal processing is needed to convert such signals into digital form for a DSP system ✓**

Answer: D. Mixed-signal processing is needed to convert such signals into digital form for a DSP system

24. Base stations, switching centres and mobile phones are given as DSP applications under which broad area?

- A. Medical imaging
- B. Satellite navigation
- C. Consumer photography
- D. Communications ✓**

Answer: D. Communications

25. GPS and GIS are cited as DSP applications belonging to which area?

- A. Measurement and control
- B. Satellite navigation ✓**
- C. Military and safety
- D. Consumer electronics

Answer: B. Satellite navigation

26. Ultrasound, MRI and CT scanning are used in the slides as DSP examples in the field of:

- A. Medical imaging and analysis ✓**
- B. Industrial measurement and control
- C. Wireless communication systems
- D. Satellite navigation and timing

Answer: A. Medical imaging and analysis

27. MP3 audio decoding, digital cameras and HDTV are grouped in the slides under which application area?

- A. Military and safety
- B. Consumer electronics ✓**
- C. Biomedical analysis
- D. Industrial control

Answer: B. Consumer electronics

28. The general problem of signal analysis and processing is framed in the course as:

- A. Letting signals pass through systems and studying the response ✓**
- B. Storing signals in memory and later retrieving them unchanged
- C. Converting each signal into one single fixed reference frequency
- D. Stripping the timing information from a signal before it is used

Answer: A. Letting signals pass through systems and studying the response

29. The general strategy the course recommends for analysing a complicated signal through a system is to:

A. Represent the signal as a sum of simple signals, get each response, then add the responses ✓

B. Measure the output for the whole complicated signal without first decomposing it

C. Replace the whole system with a single resistor-and-capacitor network before analysing

D. Convert the input into one constant value before it is applied to the system

Answer: A. Represent the signal as a sum of simple signals, get each response, then add the responses

30. A signal $f(t)$ for which $f(t) = 0$ for all $t < 0$ is called:

A. A periodic signal

B. A causal signal ✓

C. An even signal

D. A singularity signal

Answer: B. A causal signal

31. Multiplying an arbitrary (possibly non-causal) signal by the unit step $u(t)$ has the effect of:

A. Turning it into a causal signal that is zero for $t < 0$ ✓

B. Shifting it later in time by exactly one unit of time

C. Reflecting it about the vertical amplitude axis

D. Scaling its amplitude everywhere by a factor of two

Answer: A. Turning it into a causal signal that is zero for $t < 0$

32. For the sinusoid $f(t) = A \sin(2\pi f t + \phi)$, the quantity $\omega = 2\pi f$ is called the:

A. Amplitude of the waveform

B. Start phase of the waveform in radians

C. Period of the waveform in seconds

D. Angular frequency in radians per second ✓

Answer: D. Angular frequency in radians per second

33. Why does the course classify a pure sinusoid as a non-causal signal?

A. Its amplitude decays steadily toward zero as time increases

B. Being periodic, it has no start and no end in time ✓

C. It is defined for negative values of time but not positive ones

D. It changes its frequency continuously as time goes on

Answer: B. Being periodic, it has no start and no end in time

34. For the real exponential $f(t) = e^{\alpha t}$, the sign of α determines that the signal is:

A. Growing when $\alpha < 0$, and decaying when $\alpha > 0$, for any real α

B. Periodic for each real value that α can happen to take on

C. Decaying when $\alpha < 0$, constant when $\alpha = 0$, and growing when $\alpha > 0$ ✓

D. Constant at zero unless α happens to equal exactly one

Answer: C. Decaying when $\alpha < 0$, constant when $\alpha = 0$, and growing when $\alpha > 0$

35. For a complex exponential with $\alpha = \sigma + j\omega$, the real part σ governs the envelope so that the signal is:

A. A pure sinusoid when $\sigma > 0$, and a constant amplitude when $\sigma < 0$

B. A constant-amplitude wave for each value that σ may happen to take

C. A pure sinusoid when $\sigma = 0$, growing when $\sigma > 0$, and decaying (damped) when $\sigma < 0$ ✓

D. A non-oscillatory signal for whatever value the frequency ω happens to take on

Answer: C. A pure sinusoid when $\sigma = 0$, growing when $\sigma > 0$, and decaying (damped) when $\sigma < 0$

36. A gate (rectangular) signal of width τ can be written using unit step signals as:

- A. The product of two shifted unit steps taken over the interval
- B. The difference of two shifted unit steps, $u(t + \tau/2) - u(t - \tau/2)$ ✓**
- C. The integral of a single unit step evaluated over all of time
- D. The derivative of a single unit step taken right at the origin

Answer: B. The difference of two shifted unit steps, $u(t + \tau/2) - u(t - \tau/2)$

37. Why is the unit impulse $\delta(t)$ described as a singularity (generalized) function rather than a regular function?

- A. It takes a different finite constant value at each separate instant along the axis
- B. It cannot be given an ordinary value even at $t = 0$ and is defined only through an integral ✓**
- C. It stays non-zero over a wide but finite interval around the time origin
- D. It is identical to the unit step function at each point along the time axis

Answer: B. It cannot be given an ordinary value even at $t = 0$ and is defined only through an integral

38. Starting from a gate signal of width τ and unit area, the unit impulse $\delta(t)$ is obtained by:

- A. Shrinking the duration τ toward zero while keeping the area equal to one ✓**
- B. Widening the duration τ while holding the pulse height fixed at one
- C. Holding the duration fixed while raising the pulse height toward infinity
- D. Averaging the gate signal over one full period of the waveform

Answer: A. Shrinking the duration τ toward zero while keeping the area equal to one

39. As the width τ of the unit-area gate shrinks toward zero, its amplitude:

- A. Tends toward zero, so the impulse effectively vanishes from the signal
- B. Tends toward infinity, so the impulse cannot be defined by a regular function ✓**
- C. Stays fixed at one throughout the whole limiting process
- D. Swings between plus one and minus one without ever settling down

Answer: B. Tends toward infinity, so the impulse cannot be defined by a regular function

40. The sampling (sifting) property of the impulse follows from the facts that $\delta(t)$ is zero for t not equal to 0 and that:

- A. At $t = 0$ the other factor equals its value $f(0)$, a constant, over the impulse ✓**
- B. The impulse takes on a distinct value at each separate instant of time
- C. The impulse integrates to a value of zero across the whole time axis
- D. The impulse equals the unit step signal near the neighbourhood of the origin

Answer: A. At $t = 0$ the other factor equals its value $f(0)$, a constant, over the impulse

41. When the impulse is scaled as $A\delta(t)$, the constant A is referred to as the:

- A. Angular frequency, equal to the rate of change of the impulse
- B. Impulse intensity, equal to the area of the scaled impulse ✓**
- C. Start phase, equal to the time offset applied to the impulse
- D. Duration, equal to the width over which the impulse acts

Answer: B. Impulse intensity, equal to the area of the scaled impulse

42. The integral of the unit impulse $\delta(t)$ is:

- A. Another impulse of twice the area
- B. A decaying exponential signal
- C. A constant equal to zero everywhere
- D. The unit step signal ✓**

Answer: D. The unit step signal

43. With respect to symmetry, the unit impulse $\delta(t)$ is:

- A. An odd function, so $\delta(t) = -\delta(-t)$
- B. An even function, so $\delta(t) = \delta(-t)$ ✓**
- C. A causal function, zero for values of t below zero

D. A periodic function with some finite repetition period

Answer: B. An even function, so $\delta(t) = \delta(-t)$

44. The course shows that any signal $f(t)$ can be represented as:

A. A single scaled copy of the unit step function of time

B. A product of two sinusoids lying at two different frequencies

C. A constant term plus one single decaying exponential term

D. A sum (limit of an integral) of shifted and weighted unit impulses ✓

Answer: D. A sum (limit of an integral) of shifted and weighted unit impulses

45. In approximating $f(t)$ by narrow rectangles, the rectangle at $t = k \cdot (\delta\text{-tau})$ is represented as $f(k \cdot \delta\text{-tau}) \cdot (\delta\text{-tau}) \cdot \delta(t - k \cdot \delta\text{-tau})$. Making $\delta\text{-tau}$ smaller:

A. Decreases the accuracy of the rectangle-based approximation

B. Increases the accuracy, becoming exact as $\delta\text{-tau}$ tends to 0 ✓

C. Leaves the accuracy of the approximation unchanged at each step

D. Removes the impulses entirely from the signal representation

Answer: B. Increases the accuracy, becoming exact as $\delta\text{-tau}$ tends to 0

46. A system is linear if, for scaling and additivity, it satisfies:

A. $a \cdot f_1(t) + b \cdot f_2(t)$ produces $a \cdot y_1(t) + b \cdot y_2(t)$ ✓

B. $f(t - t_0)$ produces the shifted output $y(t - t_0)$ for any t_0

C. The output equals the input multiplied by one fixed constant gain

D. $f_1(t) \cdot f_2(t)$ produces the product $y_1(t) \cdot y_2(t)$ for any two inputs

Answer: A. $a \cdot f_1(t) + b \cdot f_2(t)$ produces $a \cdot y_1(t) + b \cdot y_2(t)$

47. A system is time-invariant if:

A. Shifting the input as $f(t - t_0)$ simply shifts the output to $y(t - t_0)$ ✓

B. Scaling the input by a factor of a scales the output up to $a \cdot y(t)$

C. Adding two separate inputs adds their two separate outputs

D. The output depends on the present value of the input alone

Answer: A. Shifting the input as $f(t - t_0)$ simply shifts the output to $y(t - t_0)$

48. Why are linear time-invariant (LTI) systems so central to this analysis method?

A. Summing the responses to all the shifted, weighted impulses in $f(t)$ works only for LTI systems ✓

B. They form the sole class of systems capable of processing periodic input signals

C. They yield a valid output even without any known impulse response function being available

D. They convert each input signal into a single pure sinusoid at their output

Answer: A. Summing the responses to all the shifted, weighted impulses in $f(t)$ works only for LTI systems

49. The response of an LTI system to the unit impulse $\delta(t)$ is given a special name and symbol:

A. The transfer function, denoted $F(j\omega)$

B. The unit step response, denoted $u(t)$

C. The unit impulse response, denoted $h(t)$ ✓

D. The system gain, denoted by the constant A

Answer: C. The unit impulse response, denoted $h(t)$

50. Why is the impulse response $h(t)$ regarded as such an important descriptor of an LTI system?

A. Because it only gives the system response to a steady constant direct-current input

B. Because it fixes the system's operating frequency without needing any input

C. Because it lets you skip knowing the input signal when finding the output

D. Since any signal is a sum of shifted, weighted impulses, $h(t)$ gives the response to any input ✓

Answer: D. Since any signal is a sum of shifted, weighted impulses, $h(t)$ gives the response to any input

51. When an input $f(t)$ passes through an LTI system with impulse response $h(t)$, the output is written $y(t) = f(t) * h(t)$, where the operation "*" denotes:

- A. Ordinary point-by-point multiplication
- B. Convolution of $f(t)$ and $h(t)$ ✓**
- C. Correlation of $f(t)$ with itself
- D. Simple addition of $f(t)$ and $h(t)$

Answer: B. Convolution of $f(t)$ and $h(t)$

52. Convolution $y(t) = f(t) * h(t)$ is built from the elementary LTI facts that $\delta(t) \rightarrow h(t)$ and that:

- A. A shifted impulse produces an unshifted, unscaled copy of $h(t)$
- B. Any impulse produces the unit step response rather than $h(t)$
- C. The impulse located at $t = 0$ alone produces a valid output
- D. A shifted, scaled impulse $a\delta(t - t_0)$ produces $a h(t - t_0)$ ✓**

Answer: D. A shifted, scaled impulse $a\delta(t - t_0)$ produces $a h(t - t_0)$

53. How does the course contrast the decomposition used in time-domain analysis with that used in frequency-domain analysis?

- A. Time-domain uses shifted, weighted impulses; frequency-domain uses a sinusoid plus harmonics ✓**
- B. Both of the approaches decompose the signal into shifted, weighted impulses in time
- C. Time-domain analysis uses sinusoids and harmonics while frequency-domain uses impulses
- D. Neither of the approaches decomposes the signal; each processes it as a single whole

Answer: A. Time-domain uses shifted, weighted impulses; frequency-domain uses a sinusoid plus harmonics

54. The Fourier series is a tool that applies specifically to:

- A. Causal signals
- B. Non-periodic signals
- C. Periodic signals ✓**
- D. Impulse signals

Answer: C. Periodic signals

55. In frequency-domain analysis, a signal is decomposed into:

- A. A set of shifted and weighted impulse functions
- B. A single decaying exponential component in time
- C. A product of two unit step functions of time
- D. A fundamental sinusoid together with its harmonics ✓**

Answer: D. A fundamental sinusoid together with its harmonics

56. The sampling function that arises in the spectrum of a periodic rectangular waveform is:

- A. $Sa(x) = \sin(x)/x$ ✓**
- B. $Sa(x) = \cos(x)/x$
- C. $Sa(x) = x \sin(x)$
- D. $Sa(x) = 1/(1 + x^2)$

Answer: A. $Sa(x) = \sin(x)/x$

57. The Fourier transform is introduced as the tool needed when a signal is:

- A. Periodic, though repeating only over a very short period of time
- B. Constant in both its time variable and its amplitude value
- C. Non-periodic, formed as a periodic signal's period tends to infinity ✓**
- D. Defined at only a single isolated instant along the time axis

Answer: C. Non-periodic, formed as a periodic signal's period tends to infinity

58. As the period T of a periodic signal is expanded toward infinity, both the line amplitude F_n and the spacing ω_0 tend to zero, which motivates:

- A. Keeping the discrete frequency lines exactly as they were
- B. Discarding all of the frequency information about the signal
- C. Introducing spectrum density instead of discrete frequency lines ✓**
- D. Treating the whole signal as a single impulse located in time

Answer: C. Introducing spectrum density instead of discrete frequency lines

59. The Fourier transform of a unit impulse $\delta(t)$ is a constant across all frequencies, which means the impulse:

- A. Contains only one single frequency component in its spectrum
- B. Has uniform frequency density and therefore an infinitely wide band ✓**
- C. Has essentially no frequency content anywhere in its spectrum
- D. Is band-limited to a narrow range of low frequencies

Answer: B. Has uniform frequency density and therefore an infinitely wide band

60. The Fourier transform of a constant (DC) signal of value 1 is:

- A. A constant density spread across all frequencies
- B. An impulse at zero frequency, $2\pi\delta(\omega)$ ✓**
- C. A pair of impulses at plus and minus ω_0
- D. A sinc-shaped spectrum centred at the origin

Answer: B. An impulse at zero frequency, $2\pi\delta(\omega)$

61. Why does the spectrum of the constant signal 1 consist of an impulse only at $\omega = 0$?

- A. A constant represents a direct-current signal, whose energy lies entirely at zero frequency ✓**
- B. A constant carries equal amounts of energy at every single frequency that is present
- C. A constant is a high-frequency signal that occupies only a very narrow band
- D. A constant places its energy at a pair of separate carrier frequencies only

Answer: A. A constant represents a direct-current signal, whose energy lies entirely at zero frequency

62. The Fourier transform of $\cos(\omega_0 t)$ consists of:

- A. A single impulse located exactly at zero frequency
- B. A uniform density spread across all of the frequencies
- C. A pair of impulses located at plus and minus ω_0 ✓**
- D. A sinc-shaped lobe centred at ω_0 by itself

Answer: C. A pair of impulses located at plus and minus ω_0

63. The linearity property of the Fourier transform states that:

- A. The transform of a product equals the product of the two transforms
- B. The transform of a sum equals the larger of the two transforms
- C. Scaling a signal in time has no effect at all on its transform
- D. The transform of $a*f_1(t) + b*f_2(t)$ is $aF_1(j\omega) + bF_2(j\omega)$ ✓**

Answer: D. The transform of $a*f_1(t) + b*f_2(t)$ is $aF_1(j\omega) + bF_2(j\omega)$

64. According to the time-shift property, delaying a signal by t_0 in the time domain corresponds in the frequency domain to:

- A. Scaling the magnitude of the whole spectrum by the delay value of t_0
- B. Sliding the entire spectrum bodily along the frequency axis by t_0 units
- C. Adding a fixed constant phase equal to t_0 radians at each frequency
- D. Multiplying the spectrum by the phase $e^{-j\omega t_0}$, the magnitude unchanged ✓**

Answer: D. Multiplying the spectrum by the phase $e^{-j\omega t_0}$, the magnitude unchanged

65. The frequency-shift (modulation) property says that multiplying $f(t)$ by $e^{j\omega_0 t}$ in the time domain:

- A. Multiplies the spectrum by the constant factor $e^{-j\omega_0}$

- B. Delays the signal by ω_0 seconds along the time axis
- C. Shifts the spectrum to $F(j(\omega - \omega_0))$ in the frequency domain ✓**
- D. Leaves the spectrum of the signal unaffected altogether

Answer: C. Shifts the spectrum to $F(j(\omega - \omega_0))$ in the frequency domain

66. Multiplying a baseband signal $f(t)$ by a carrier $\cos(\omega_0 t)$ produces a spectrum that is:

- A. A single copy of $F(j\omega)$ left sitting right at the frequency origin
- B. An impulse at zero frequency scaled by the carrier amplitude
- C. The original spectrum $F(j\omega)$ with its bandwidth halved
- D. Two half-amplitude copies of $F(j\omega)$ centred at plus and minus ω_0 ✓**

Answer: D. Two half-amplitude copies of $F(j\omega)$ centred at plus and minus ω_0

67. Parseval's relation (the energy theorem) expresses that:

- A. The total energy of a real-valued signal always works out to be zero
- B. The phase of a signal carries the whole of its energy information
- C. Energy is conserved only for perfectly periodic signals but not for others
- D. The total energy of a signal can be computed in the time or the frequency domain ✓**

Answer: D. The total energy of a signal can be computed in the time or the frequency domain

68. The quantity $|F(j\omega)|^2$ is called the signal's energy spectrum because it represents:

- A. The instantaneous power carried by the signal at each separate moment in time
- B. The phase of the signal expressed as a function of the frequency
- C. The signal energy contained in a unit band of frequency, carrying no phase information ✓**
- D. The total count of the frequency lines making up the whole spectrum

Answer: C. The signal energy contained in a unit band of frequency, carrying no phase information

69. The time-domain convolution theorem of the Fourier transform states that:

- A. Convolution in time corresponds instead to convolution of the two spectra as well
- B. Convolution in time corresponds to simple addition of the two spectra
- C. Convolution in time leaves each of the two signal spectra unchanged
- D. Convolution in time corresponds to multiplication of the spectra, $F_1(j\omega)F_2(j\omega)$ ✓**

Answer: D. Convolution in time corresponds to multiplication of the spectra, $F_1(j\omega)F_2(j\omega)$

70. The frequency-domain convolution theorem states that multiplication of two signals in time corresponds in frequency to:

- A. Convolution of their spectra, scaled by a factor of $1/(2\pi)$ ✓**
- B. Multiplication of their spectra with no extra scaling factor
- C. Addition of their two spectra with a factor of 2π applied
- D. The difference of their two spectra taken across frequency

Answer: A. Convolution of their spectra, scaled by a factor of $1/(2\pi)$

71. What broad usefulness does the convolution theorem provide, according to the slides?

- A. It converts an input signal into a single sharp impulse located firmly in time
- B. It removes the need to know a system's impulse response when solving
- C. It applies solely to signals that happen to be perfectly periodic ones
- D. It turns convolution in one domain into an algebraic operation in the other domain ✓**

Answer: D. It turns convolution in one domain into an algebraic operation in the other domain

72. For a linear system, the quantity $H(j\omega)$ that relates input and output spectra is called the:

- A. Energy spectrum of the applied input signal in frequency
- B. Impulse intensity of the system
- C. System function, also called the transfer function ✓**
- D. Sampling function of the output

Answer: C. System function, also called the transfer function

73. If a system has impulse response $h(t)$ and transfer function $H(j\omega)$, then the output spectrum is given by:

- A. $Y(j\omega) = F(j\omega) \cdot H(j\omega)$, the product of input spectrum and transfer function ✓
- B. $Y(j\omega) = F(j\omega) + H(j\omega)$, the sum of the input and the function
- C. $Y(j\omega) = F(j\omega) / H(j\omega)$, the ratio of the input to the transfer function
- D. $Y(j\omega) = F(j\omega)$, taken independently of the system's own function

Answer: A. $Y(j\omega) = F(j\omega) \cdot H(j\omega)$, the product of input spectrum and transfer function

74. The fact that $Y(j\omega) = F(j\omega) \cdot H(j\omega)$ in the frequency domain corresponds in the time domain to:

- A. The product $y(t) = f(t) \cdot h(t)$ taken sample by sample in time
- B. The sum $y(t) = f(t) + h(t)$ taken point by point in time
- C. The convolution $y(t) = f(t) * h(t)$ taken in the time domain ✓
- D. The derivative of $f(t)$ with respect to time at each instant

Answer: C. The convolution $y(t) = f(t) * h(t)$ taken in the time domain

75. If a system is too complicated to express analytically, the slides suggest measuring its transfer function by:

- A. Applying sine inputs at different frequencies, measuring each output, and combining them ✓
- B. Applying a single impulse and reading off the final steady-state value
- C. Applying a constant direct-current input and then scaling up the measured result
- D. Computing the ratio between two arbitrarily chosen samples in time

Answer: A. Applying sine inputs at different frequencies, measuring each output, and combining them

76. The technique of driving a system with sine inputs across a range of frequencies to characterise it is called:

- A. A convolution sum
- B. A frequency sweep ✓
- C. A time reversal
- D. An impulse decomposition

Answer: B. A frequency sweep

77. Euler's relation is invoked in this material chiefly to:

- A. Express sine and cosine signals in terms of complex exponentials ✓
- B. Convert a periodic signal into an equivalent causal signal
- C. Prove that the unit impulse is an even function of time
- D. Define the sampling interval used for a discrete signal

Answer: A. Express sine and cosine signals in terms of complex exponentials

78. The Fourier transform of the one-sided exponential $e^{-a t} u(t)$ with $a > 0$ has a magnitude spectrum that:

- A. Is a pair of sharp impulses located at plus and minus a
- B. Is flat and equal in value at every one of the frequencies
- C. Is zero at the low frequencies and then rises without any upper bound
- D. Is largest at zero frequency and then rolls off as the frequency increases ✓

Answer: D. Is largest at zero frequency and then rolls off as the frequency increases

79. In the time domain, a discrete-time signal is represented as:

- A. A continuous waveform defined at every instant
- B. A single scalar value that never changes in time
- C. A frequency spectrum holding no time information
- D. A sequence of numbers, called samples ✓

Answer: D. A sequence of numbers, called samples

80. When a discrete-time sequence is formed by sampling a continuous signal $x_a(t)$, the n -th

sample is given by:

- A. $x[n] = x_a(t)$ integrated from time 0 up to $t = nT$
 - B. $x[n] = x_a(nT)$, the value of $x_a(t)$ taken at $t = nT$ ✓**
 - C. $x[n] = x_a(t)$ averaged across the interval of width T
 - D. $x[n] =$ the derivative of $x_a(t)$ evaluated at $t = nT$
- Answer: B. $x[n] = x_a(nT)$, the value of $x_a(t)$ taken at $t = nT$**

81. The spacing T between two consecutive samples is called the sampling interval, and its reciprocal $1/T$ is called the:

- A. Angular frequency of the samples
- B. Fundamental period of the sequence
- C. Quantization step of the sequence
- D. Sampling frequency ✓**

Answer: D. Sampling frequency

82. The course distinguishes sampled-data signals from digital signals by noting that:

- A. Sampled-data signals have continuous-valued samples, while digital signals have discrete-valued samples ✓**
- B. Sampled-data signals have discrete-valued samples, while the digital ones are continuous-valued
- C. Both sampled-data and digital signals turn out to have continuous-valued samples
- D. Sampled-data signals alone are the ones actually defined at discrete time instants

Answer: A. Sampled-data signals have continuous-valued samples, while digital signals have discrete-valued samples

83. Signals in a practical DSP system are digital signals, obtained from sample values by:

- A. Convolution with the unit step sequence in time
- B. Quantizing them, either by rounding or by truncation ✓**
- C. Reversing them in time about the sampling origin
- D. Interpolating them up to a higher sampling rate

Answer: B. Quantizing them, either by rounding or by truncation

84. A right-sided sequence has zero-valued samples for $n < N_1$. It is additionally called a causal sequence when:

- A. $N_1 < 0$
- B. $N_1 \geq 0$ ✓**
- C. N_1 is a non-integer value
- D. N_1 tends to negative infinity

Answer: B. $N_1 \geq 0$

85. In the modulation (product) operation on sequences, the output is formed by:

- A. Adding the two input sequences sample by sample, $y[n] = x[n] + w[n]$
- B. Delaying one of the input sequences by a single sampling interval
- C. Multiplying the two input sequences sample by sample, $y[n] = x[n] \cdot w[n]$ ✓**
- D. Reversing one input sequence in time before they are combined

Answer: C. Multiplying the two input sequences sample by sample, $y[n] = x[n] \cdot w[n]$

86. The scalar-multiplication operation on a sequence produces:

- A. $y[n] = A \cdot x[n]$, each sample scaled by a constant A ✓**
- B. $y[n] = x[n] + A$, a constant A added to each sample
- C. $y[n] = x[n - A]$, the sequence delayed by A samples
- D. $y[n] = x[A \cdot n]$, the sequence resampled in the time index

Answer: A. $y[n] = A \cdot x[n]$, each sample scaled by a constant A

87. A unit-delay operation applied to $x[n]$ produces the output:

- A. $y[n] = x[n + 1]$
- B. $y[n] = x[-n]$

C. $y[n] = x[n - 1]$ ✓

D. $y[n] = A \cdot x[n]$

Answer: C. $y[n] = x[n - 1]$

88. The time-reversal (folding) operation on a sequence produces:

A. $y[n] = x[n - 1]$

B. $y[n] = x[n] + x[-n]$

C. $y[n] = x[-n]$ ✓

D. $y[n] = x[n] \cdot x[-n]$

Answer: C. $y[n] = x[-n]$

89. The branching operation in a discrete-time structure is used to:

A. Combine several separate sequences into just one single output

B. Delay a sequence by exactly one sampling period in time

C. Provide multiple copies of a sequence for use at different points ✓

D. Reverse a sequence about the origin of the time axis

Answer: C. Provide multiple copies of a sequence for use at different points

90. In sampling-rate alteration, the ratio $R = F'_T / F_T$ describes the change of rate. The process is called interpolation when:

A. $R < 1$, giving a lower sampling rate

B. $R = 1$, leaving the rate unchanged

C. R is negative, reversing the sequence

D. $R > 1$, giving a higher sampling rate ✓

Answer: D. $R > 1$, giving a higher sampling rate

91. In sampling-rate alteration, the process is called decimation when:

A. $R > 1$, giving a higher sampling rate

B. $R = 0$, removing all of the samples

C. $R < 1$, giving a lower sampling rate ✓

D. $R = 1$, leaving the rate unchanged

Answer: C. $R < 1$, giving a lower sampling rate

92. A discrete-time sequence is classified as periodic when:

A. Its samples all settle to one and the same constant value

B. It has only a finite number of non-zero samples in total

C. Its samples decay steadily toward zero as n increases

D. It repeats, so that $x[n]$ equals $x[n + N]$ for some period N ✓

Answer: D. It repeats, so that $x[n]$ equals $x[n + N]$ for some period N

93. With respect to length and energy, which statement matches the course's classification?

A. A finite-length sequence with finite sample values instead has infinite energy

B. An infinite-length sequence is guaranteed to end up with finite energy

C. A sequence that has finite energy is bound to have an infinite length

D. A finite-length sequence with finite sample values always has finite energy ✓

Answer: D. A finite-length sequence with finite sample values always has finite energy

94. A sequence is classified as a power signal when it has:

A. Infinite energy but finite average power, as a periodic sequence has ✓

B. Finite energy but zero average power, as a short sequence has

C. Both finite energy and finite average power at the same time

D. Zero energy together with zero average power everywhere

Answer: A. Infinite energy but finite average power, as a periodic sequence has

95. A sequence is classified as an energy signal when it has:

A. Infinite energy but finite average power, as a periodic sequence has

- B. Infinite energy together with an infinite average power at once
 - C. Finite energy together with a finite, non-zero average power
 - D. Finite energy but zero average power, as a finite-length sequence has ✓**
- Answer: D. Finite energy but zero average power, as a finite-length sequence has**

96. A sequence $x[n]$ is said to be bounded when:

- A. Its magnitude stays below a finite limit for all n , as $\cos(0.3\pi n)$ does ✓**
 - B. The sum of all of its sample values turns out to be finite
 - C. The sum of the squares of all of its sample values turns out finite
 - D. It has only a finite number of non-zero samples in total
- Answer: A. Its magnitude stays below a finite limit for all n , as $\cos(0.3\pi n)$ does**

97. The two fundamental basic sequences introduced first in the discrete-time material are:

- A. The gate sequence and the linear ramp sequence
 - B. The chirp sequence and the constant sequence
 - C. The random sequence and the periodic sequence
 - D. The unit sample sequence and the unit step sequence ✓**
- Answer: D. The unit sample sequence and the unit step sequence**

98. A real sinusoidal sequence is written $x[n] = A\cos(\omega_0 n + \phi)$, where the symbol ω_0 is the:

- A. Angular frequency of the sequence ✓**
 - B. Amplitude of the sequence
 - C. Phase offset of the sequence
 - D. Sampling interval of the sequence
- Answer: A. Angular frequency of the sequence**

99. For a complex exponential sequence, the real part σ_0 of the exponent controls the amplitude so that the real sinusoidal parts are:

- A. Constant when $\sigma_0 = 0$, growing when $\sigma_0 > 0$, and decaying when $\sigma_0 < 0$ ✓**
 - B. Constant when $\sigma_0 > 0$, and growing when σ_0 turns out to equal zero
 - C. Growing whenever σ_0 is less than zero, for any amplitude
 - D. Held at zero unless σ_0 turns out to equal exactly one
- Answer: A. Constant when $\sigma_0 = 0$, growing when $\sigma_0 > 0$, and decaying when $\sigma_0 < 0$**

100. A sinusoidal sequence $A\cos(\omega_0 n + \phi)$ is periodic only when:

- A. ω_0 is any real number that is greater than zero
 - B. ω_0 equals the sampling frequency of the sequence exactly
 - C. $\omega_0 N = 2\pi r$ for positive integers N and r ✓**
 - D. N is chosen freely as any convenient non-integer value
- Answer: C. $\omega_0 N = 2\pi r$ for positive integers N and r**

101. For a periodic sequence, the fundamental period N is:

- A. The largest value of N that satisfies the periodicity condition
 - B. The smallest value of N satisfying $\omega_0 N = 2\pi r$ ✓**
 - C. Any value of N for which $\omega_0 N$ is a whole integer
 - D. A value equal to the amplitude A of the sinusoidal sequence
- Answer: B. The smallest value of N satisfying $\omega_0 N = 2\pi r$**

102. The sampling theorem (Nyquist theorem) states that a continuous signal can be represented uniquely by its samples if the sampling frequency is:

- A. Equal to the highest frequency that is contained in the signal
 - B. Less than half of the highest frequency present in the signal
 - C. Set independently of which frequencies are present in the signal
 - D. Greater than twice the highest frequency contained in the signal ✓**
- Answer: D. Greater than twice the highest frequency contained in the signal**

103. The specific problem that the sampling theorem is meant to prevent is:

- A. Quantization error, which grows when too few bits are used
- B. Truncation, which cuts the sequence down to a finite length
- C. Phase loss, which discards the timing detail of the signal
- D. Aliasing, which occurs when the sampling frequency is too low ✓**

Answer: D. Aliasing, which occurs when the sampling frequency is too low

104. A discrete-time system is defined to be linear when, for $x[n] = a \cdot x_1[n] + b \cdot x_2[n]$, the output is:

- A. $y[n] = y_1[n - n_0]$, a shifted version of one of the responses
- B. $y[n] = y_1[n] \cdot y_2[n]$, the product of the two responses formed
- C. $y[n] = a \cdot y_1[n] + b \cdot y_2[n]$ for any constants a and b and any inputs ✓**
- D. $y[n] = a \cdot y_1[n]$ on its own, disregarding the second input

Answer: C. $y[n] = a \cdot y_1[n] + b \cdot y_2[n]$ for any constants a and b and any inputs

105. A discrete-time system is shift-invariant when a shift of the input by n_0 , so that $x[n] = x_1[n - n_0]$, produces:

- A. $y[n] = y_1[n - n_0]$, the same response shifted along by n_0 ✓**
- B. $y[n] = y_1[n]$, the same response with no shift applied to it
- C. $y[n] = n_0 \cdot y_1[n]$, the response scaled by the factor n_0
- D. $y[n] = y_1[-n]$, the response reversed about the time origin

Answer: A. $y[n] = y_1[n - n_0]$, the same response shifted along by n_0

106. A linear time-invariant (LTI) discrete-time system is one that:

- A. Satisfies linearity but fails the shift-invariance property
- B. Satisfies shift-invariance but fails the linearity property
- C. Satisfies both the linearity and the shift-invariance properties ✓**
- D. Satisfies neither property yet remains causal in all cases

Answer: C. Satisfies both the linearity and the shift-invariance properties

107. Why are LTI discrete-time systems emphasised so strongly in the course?

- A. They form the sole class that is able to process periodic input sequences properly
- B. They produce an output without any impulse response being known
- C. They require the sampling rate to be altered before they are used
- D. They are easy to analyze, characterize and design, and support many useful algorithms ✓**

Answer: D. They are easy to analyze, characterize and design, and support many useful algorithms

108. The response of a discrete-time system to a unit sample sequence $\delta[n]$ is called the:

- A. Unit impulse response, denoted $h[n]$ ✓**
- B. Unit step response, denoted $s[n]$
- C. Transfer function, denoted by H
- D. Convolution sum, denoted $y[n]$

Answer: A. Unit impulse response, denoted $h[n]$

109. A key consequence of the linear, time-invariant property is that an LTI discrete-time system is:

- A. Characterized instead by its unit step response
- B. Independent of whatever input is applied to it
- C. Fully described by a single output sample value
- D. Completely characterized by its impulse response ✓**

Answer: D. Completely characterized by its impulse response

110. Because any input can be written as a linear combination of shifted unit samples, the response of an LTI system to an input $x[k] \cdot \delta[n - k]$ is:

- A. $x[k] \cdot \delta[n - k]$, left unchanged by the system
- B. $x[k] \cdot h[n - k]$ ✓**

- C. $h[n]$, taken independently of the index k
- D. $x[n] * h[k]$, with the two indices swapped over

Answer: B. $x[k] * h[n - k]$

111. The summation that combines $x[n]$ and $h[n]$ to give the output of an LTI system, written $y[n] = x[n] * h[n]$, is called the:

- A. Fourier series
- B. Convolution sum ✓**
- C. Frequency sweep
- D. Sampling sum

Answer: B. Convolution sum

112. The commutative property of the convolution sum states that:

- A. $x[n] * h[n] = x[n] + h[n]$
- B. $x[n] * h[n] = x[-n] * h[-n]$
- C. $x[n] * h[n]$ varies with the operand order
- D. $x[n] * h[n] = h[n] * x[n]$ ✓**

Answer: D. $x[n] * h[n] = h[n] * x[n]$

113. The distributive property of the convolution sum states that:

- A. $x[n] * (h[n] + y[n]) = x[n] * h[n] + x[n] * y[n]$ ✓**
- B. $x[n] * (h[n] + y[n]) = x[n] * h[n]$ on its own
- C. $x[n] * (h[n] * y[n]) = (x[n] * h[n]) * y[n]$
- D. $x[n] * (h[n] + y[n]) = h[n] + y[n]$ regardless of x

Answer: A. $x[n] * (h[n] + y[n]) = x[n] * h[n] + x[n] * y[n]$

114. In the graphical interpretation of the convolution sum, the impulse response $h[k]$ is first:

- A. Scaled by a fixed constant, then integrated over all of the values of k
- B. Left just as it is, then added straight onto the sequence $x[k]$
- C. Time-reversed to form $h[-k]$, then shifted before multiplying and summing ✓**
- D. Resampled to a higher rate, then quantized before being summed

Answer: C. Time-reversed to form $h[-k]$, then shifted before multiplying and summing

115. If two sequences of lengths M and N are convolved, the resulting sequence has length:

- A. $M + N$
- B. $M + N - 1$ ✓**
- C. $M * N$
- D. the larger of M and N

Answer: B. $M + N - 1$

116. A discrete-time LTI system is classified as a finite impulse response (FIR) system when its impulse response $h[n]$ is:

- A. Of infinite length
- B. Of finite length ✓**
- C. Equal to the unit step sequence
- D. Undefined for negative values of n

Answer: B. Of finite length

117. A discrete-time LTI system is classified as an infinite impulse response (IIR) system when its impulse response is:

- A. Of finite length
- B. Zero outside one single sample
- C. Of infinite length ✓**
- D. Identical to its own input sequence

Answer: C. Of infinite length

118. The IIR systems studied in this course are characterized mathematically by:

- A. Finite sums of products with no feedback path
- B. Linear constant-coefficient difference equations ✓**
- C. A single scalar-multiplication operation
- D. The Fourier series of a periodic signal

Answer: B. Linear constant-coefficient difference equations

119. The discrete-time accumulator defined by $y[n] = y[n - 1] + x[n]$ is given as an example of:

- A. An FIR system, since it uses present and past inputs only
- B. A non-linear system, since it adds two signals together
- C. An IIR system, since its impulse response is of infinite length ✓**
- D. A time-varying system, since its coefficients change over time

Answer: C. An IIR system, since its impulse response is of infinite length

120. A discrete-time system is called non-recursive when its output can be computed:

- A. Using past output samples along with the inputs
- B. Without knowing any of the input samples at all
- C. From just the present and past input samples ✓**
- D. Once the whole input sequence has been received

Answer: C. From just the present and past input samples

121. A discrete-time system is called recursive when its output computation:

- A. Uses just the single present input sample each time
- B. Uses present and past input samples but no outputs
- C. Involves past output samples besides present and past inputs ✓**
- D. Leaves out all the earlier values of the input and the output

Answer: C. Involves past output samples besides present and past inputs

122. The response of a discrete-time system to a unit step sequence is called the:

- A. Unit impulse response, denoted $h[n]$
- B. Convolution sum, denoted $y[n]$
- C. Transfer function, denoted $H[n]$
- D. Unit step response, denoted $s[n]$ ✓**

Answer: D. Unit step response, denoted $s[n]$

123. An arbitrary discrete-time sequence can be represented in the time domain as:

- A. A single scaled copy of the unit step sequence in time
- B. The product of two sinusoidal sequences of different rates
- C. A weighted sum of a basic sequence and its shifted versions ✓**
- D. A constant term plus one single decaying exponential term

Answer: C. A weighted sum of a basic sequence and its shifted versions

124. For a complex exponential sequence written $A \cdot \alpha^n$, the symbols A and alpha are, in general:

- A. Strictly positive integers
- B. Real or complex numbers ✓**
- C. Numbers equal to one another
- D. Values between zero and one

Answer: B. Real or complex numbers